

TWENTY NINE PALMS, CA, USA

WMO: **690150**Lat: **34.300N**Lon: **116.167W**Elev: **2051**StdP: **13.64**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **93121**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.0	34.2	-5.3	4.5	51.7	-1.2	5.5	50.7	33.8	49.1	28.9	50.2	4.3	310	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	108.3	68.2	105.8	67.3	103.3	66.1	74.2	97.7	72.3	95.3	70.9	94.3	9.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	111.4	82.9	65.0	100.0	84.8	62.4	91.0	87.4	38.9	97.8	37.2	94.8	35.9	94.1	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	24.4	20.3	DB	24.6	112.7	4.1	3.6	21.7	115.3	19.3	117.3	17.0	119.3	14.1	121.9	(4)
(5)			WB	19.2	74.9	3.4	2.0	16.7	76.4	14.7	77.5	12.7	78.6	10.2	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	50.6	53.9	60.8	66.6	75.0	84.9	90.2	89.1	82.2	69.7	58.2	49.6		
	DBStd	15.66	6.03	6.53	6.79	7.51	7.56	6.36	4.65	4.16	6.56	6.60	6.78	5.80		
	HDD50	198	67	32	4	1	0	0	0	0	0	0	14	80		
	HDD65	1739	447	313	167	70	13	0	0	0	0	33	218	478		
	CDD50	7240	85	141	337	497	774	1046	1247	1213	965	609	259	67		
	CDD65	3309	0	2	35	117	322	596	782	748	516	177	14	0		
	CDH74	48124	3	67	490	1470	4016	8892	12408	11557	6968	1996	255	2		
	CDH80	29391	0	5	109	568	2083	5666	8334	7579	4156	845	46	0		
(14) Wind	WSAvg	8.1	6.8	7.8	8.9	10.5	10.4	9.2	8.4	7.7	6.9	6.9	6.6	7.0		
(15) Precipitation	PrecAvg	4.7	1.1	0.9	0.5	0.2	0.0	0.0	0.3	0.3	0.4	0.2	0.4	0.6		
	PrecMax	10.2	6.5	3.1	2.5	0.9	0.3	0.2	1.9	1.1	3.2	1.5	1.6	3.1		
	PrecMin	1.7	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.4	1.5	0.9	0.6	0.2	0.1	0.1	0.4	0.3	0.6	0.4	0.4	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.6	79.8	87.7	95.0	102.6	111.4	111.3	109.4	113.0	96.2	84.3	73.5		
		MCWB	49.9	53.0	56.3	60.3	63.9	66.5	70.0	70.4	67.6	62.8	56.0	51.1		
	2%	DB	70.2	75.8	82.8	90.5	98.4	106.3	108.4	106.8	102.4	91.6	80.6	69.5		
		MCWB	48.3	51.7	54.4	58.3	61.6	64.9	68.9	68.3	65.9	60.3	53.8	49.6		
	5%	DB	66.9	71.9	79.6	86.5	94.8	103.3	105.9	104.4	99.5	88.4	76.6	65.6		
		MCWB	47.2	50.2	53.2	56.5	59.6	63.8	68.1	67.5	66.3	58.9	52.7	47.5		
	10%	DB	63.3	67.6	76.0	82.3	91.3	100.4	103.1	102.1	97.1	84.7	72.2	61.9		
		MCWB	46.3	48.6	51.7	54.9	58.6	63.6	66.9	67.2	64.6	57.1	51.2	46.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	56.8	57.7	63.1	68.3	73.0	77.9	78.7	73.5	66.4	58.7	55.4		
		MCDB	62.9	72.6	82.0	89.3	99.0	103.1	100.8	100.8	98.1	84.9	76.9	67.0		
	2%	WB	51.4	53.8	55.5	59.9	63.7	69.5	74.0	74.7	71.1	63.8	55.9	52.8		
		MCDB	62.9	69.3	79.0	87.1	93.1	100.5	97.3	97.7	94.8	84.4	76.5	62.7		
	5%	WB	49.7	51.8	54.2	57.5	61.2	67.2	72.2	72.6	69.5	61.3	53.8	50.2		
		MCDB	62.3	66.9	76.9	83.3	90.2	99.1	94.8	94.7	92.1	84.5	73.2	61.5		
	10%	WB	47.6	50.0	52.6	55.4	59.3	64.9	70.8	70.9	67.9	59.1	51.8	47.9		
		MCDB	61.6	64.9	73.9	79.9	88.4	96.1	94.0	93.6	89.8	81.9	70.5	60.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.7	23.3	24.8	25.6	25.7	27.7	24.9	25.6	26.3	26.3	24.7	22.5		
		MCDBR	27.4	28.6	28.7	29.6	28.8	29.3	26.5	27.5	28.4	30.6	29.8	27.7		
	5% WB	MCWBR	14.3	13.5	12.5	13.0	10.5	11.0	10.6	11.2	11.9	14.2	15.0	14.6		
		MCWBR	23.1	25.2	26.9	28.1	27.6	28.1	22.9	23.5	24.5	28.1	26.8	23.2		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.277	0.291	0.316	0.327	0.324	0.359	0.352	0.325	0.302	0.276	0.275		
	taud		2.550	2.557	2.502	2.422	2.420	2.443	2.429	2.433	2.485	2.522	2.574	2.567		
	Ebn at Noon		302	312	313	307	303	302	291	292	297	298	298	295		
	Edh at Noon		25	28	32	36	37	36	36	36	32	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		1055	1369	1838	2260	2540	2698	2436	2277	2000	1564	1171	953		
	RadStd		70	109	74	66	87	96	112	94	71	69	48	57		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	+1.04	N/A	-3.08	+1.04	N/A	N/A	N/A	N/A	-164	+367	+202

Nomenclature: See separate page